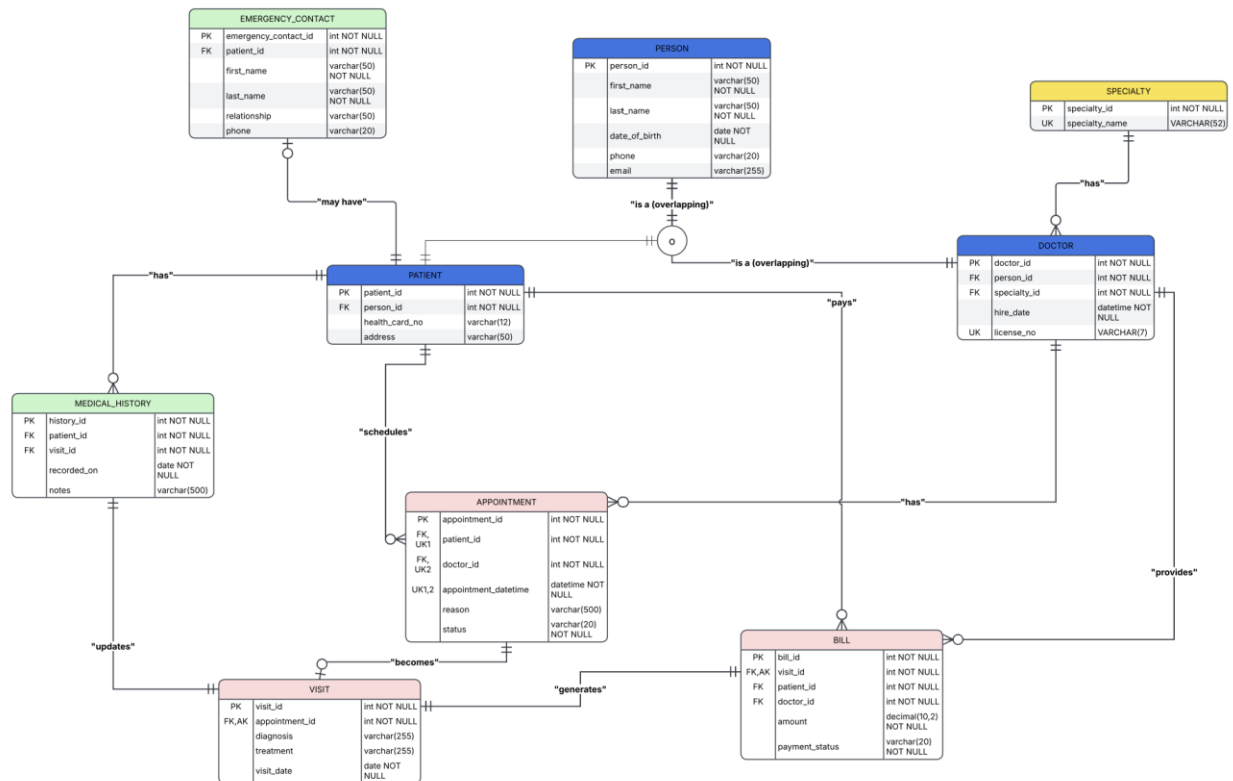


Class Activity: Implementing and Querying a Relational Database

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EERD



Main Queries:

Kyle

1. Count the total number of patients in the database.

total_patients	
bigint	
1	6

This query counts the total number of patients in the database

2. Count how many patients visited the clinic last month.

	patients_visited_last_month bigint	
1		3

This query counts how many patients visited the clinic in the past month

3. Calculate the total dollar amount billed to all patients.

	total_billed numeric	
1	1780.50	

This query calculates the total amount that was billed to all patients in the database

Custom Queries

Ethan:

--- 1. Get total bills, total sum of bills, and average \$ value

	total_bills bigint	total_billing numeric	average_bill numeric	
1	11	1780.50	161.86	

This query returns the total bills of patients in the database, along with sum of those bills, as well as the average amount of the bills

--- 2. Subqueries and JOINS: Get all patients, the total amount they've been billed, and their doctors name

	patient_id integer	first_name character varying (50)	last_name character varying (50)	total_billed numeric	doctor_first_name character varying (50)	doctor_last_name character varying (50)
1	1	James	Anderson	645.00	Sarah	Mitchell
2	2	Maria	Garcia	320.50	Priya	Sharma
3	3	Liam	Chen	250.00	David	Nguyen
4	4	Aisha	Khan	210.00	Michael	Okafor
5	5	Robert	Tremblay	160.00	Elena	Rossi
6	6	Olivia	Brown	195.00	Elena	Rossi

This query joins the tables Patients, Bills, and Doctors to return a query that shows patients names, what they were billed and who the doctor was

--- 3. GROUP BY with HAVING: Get doctors with over \$200 total billed

	doctor_id integer	first_name character varying (50)	last_name character varying (50)	total_billed numeric
1	5	Elena	Rossi	355.00
2	1	Sarah	Mitchell	645.00
3	4	Michael	Okafor	210.00
4	2	David	Nguyen	250.00
5	3	Priya	Sharma	320.50

This query retrieves all doctors that have billed over 200\$ within their employment at SAIT's clinic.

Drew

--- 4. View: doctor billing summary

	doctor_id integer	first_name character varying (50)	last_name character varying (50)	total_bills bigint	total_billed numeric	average_bill numeric
1	1	Sarah	Mitchell	3	645.00	215.00
2	5	Elena	Rossi	3	355.00	118.33
3	3	Priya	Sharma	2	320.50	160.25
4	2	David	Nguyen	2	250.00	125.00
5	4	Michael	Okafor	1	210.00	210.00

This query displays a view of each doctor's total bill amount, and average bill amount, summarizing doctor productivity.

--- 5. For each doctor, who are their two highest-billed patients?

	doctor_id integer	doctor_first_name character varying (50)	doctor_last_name character varying (50)	patient_id integer	patient_first_name character varying (50)	patient_last_name character varying (50)	total_billed numeric	patient_rank bigint
1	1	Sarah	Mitchell	1	James	Anderson	645.00	1
2	2	David	Nguyen	3	Liam	Chen	250.00	1
3	3	Priya	Sharma	2	Maria	Garcia	320.50	1
4	4	Michael	Okafor	4	Aisha	Khan	210.00	1
5	5	Elena	Rossi	6	Olivia	Brown	195.00	1
6	5	Elena	Rossi	5	Robert	Tremblay	160.00	2

This query displays the two highest billed patients for each doctor. Unfortunately, our data set was small, thus doctors had only one unique patient billed.

Brayden

--- 6. How did visits and billing change from month to month?

```
26 FROM monthly_changes
27 ORDER BY visit_month;
```

	visit_month text	total_visits bigint	total_billing numeric	visit_change bigint	billing_change numeric
1	2026-05	2	350.00	[null]	[null]
2	2026-06	1	160.00	-1	-190.00
3	2026-07	3	515.00	2	355.00
4	2026-08	2	430.50	-1	-84.50
5	2026-09	3	325.00	1	-105.50

This query shows how the visits and changes on a month-to-month basis. This is important for showing the business traffic and whether profits are up or down compared to last month.

--- 7. Doctors who bill more than the average doctor, and their share of all clinic billing

```
31 WHERE total_billed > (SELECT AVG(total_billed) FROM doctor_totals)
32 ORDER BY total_billed DESC;
```

	doctor_id integer	first_name character varying (50)	last_name character varying (50)	total_billed numeric	clinic_billing_share_pct numeric
1	1	Sarah	Mitchell	645.00	36.23

This query shows the contribution of doctors who are making more money than the clinic average and displaying their percentage of earnings in contrast to the clinic.

Team Contributions

Name	Contribution
Brayden	EERD, Query 6, 7, Documentation
Drew	EERD, Query 4,5, Documentation
Ethan	Create Table Schema, Query 1,2,3
Kyle	Create Table Schema, Main Queries 1,2,3